

SCHEMATIC PAGE DESCRIPTION :

PAGE01: BLOCK DIAGRAM

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PAGE04: FPGA BANK15&16

PAGE05: FPGA BANK34&35

PAGE06: FPGA BANK216

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PAGE08: FPGA Power

PAGE09: DDR3

PAGE10: System Power

PAGE11: IO Connectors

JTAG

IO Headers

CLOCK

BLOCK DIAGRAM

FPGA

Migration Devices:
XC7A35T-Xxxx484
XC7A50T-Xxxx484
XC7A75T-Xxxx484
XC7A100T-Xxxx484
XC7A200T-Xxxx484

USB UART

DDR3

LEDS

KEY

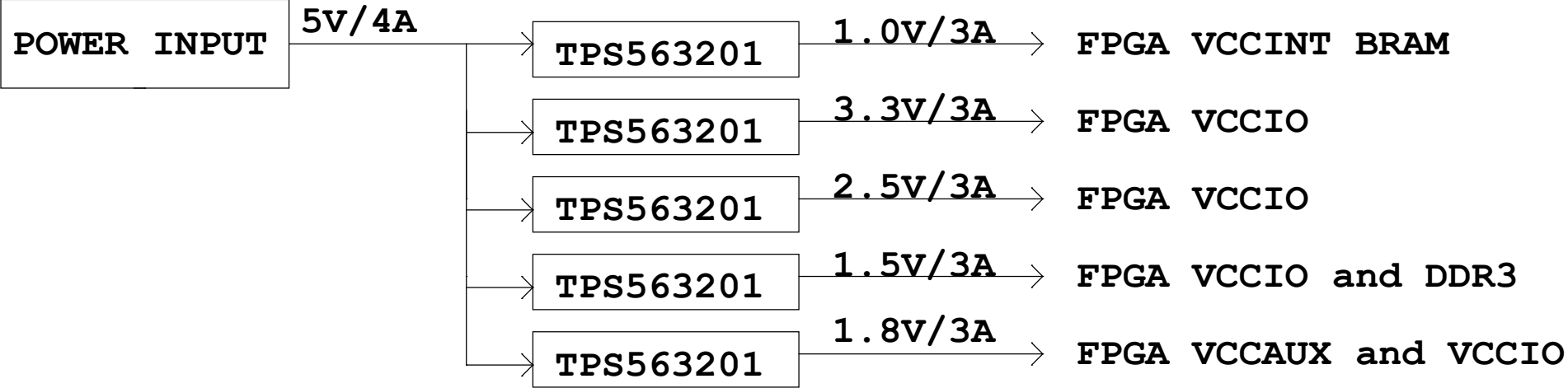
Config FLASH



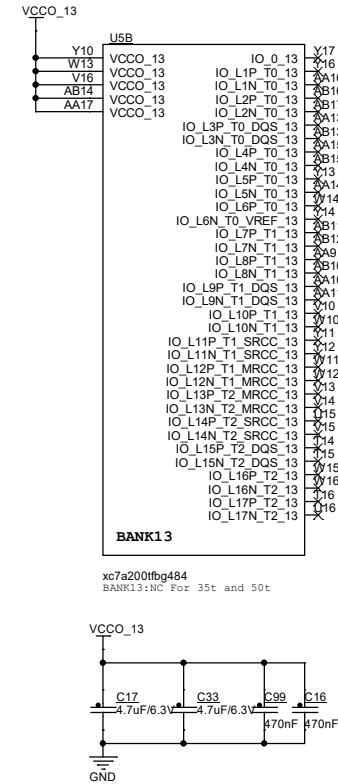
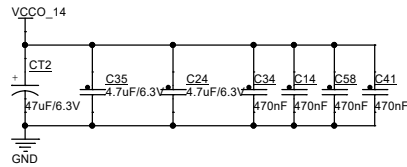
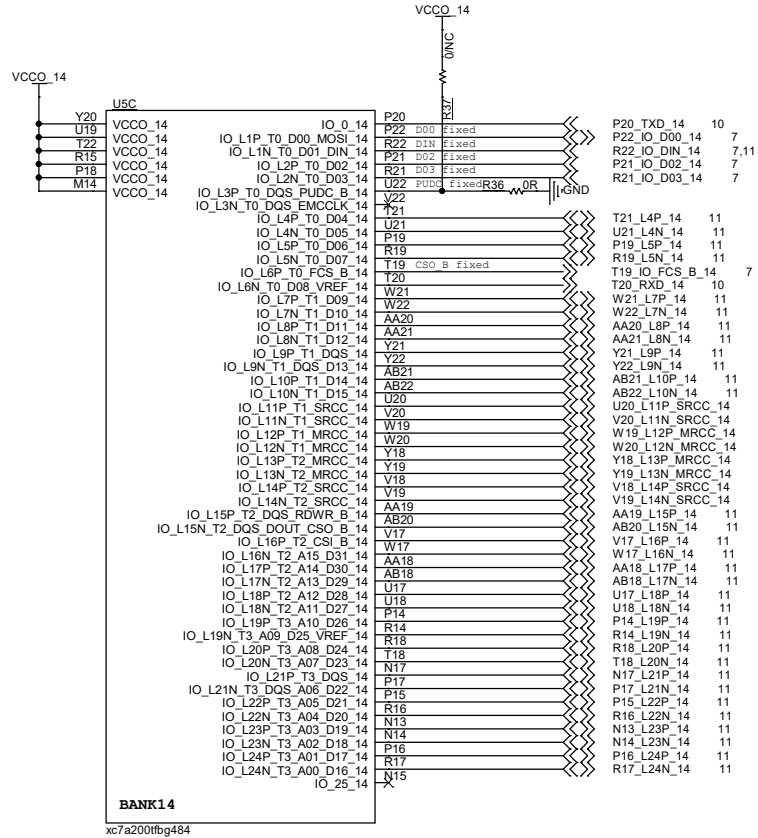
AGILCIR

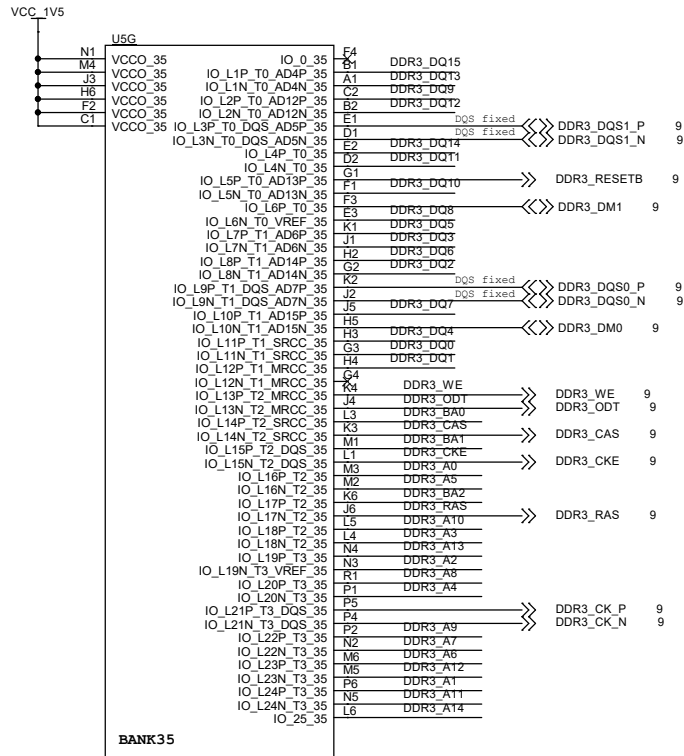
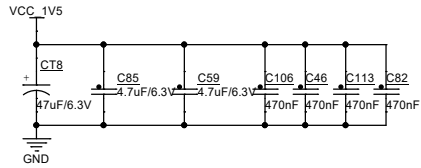
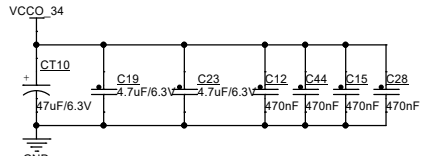
Chengdu Heiying electronic technology co.,Ltd			
Title		Xilinx Artix-7 FPGA A704	
Size	Document	Number	Rev
A3	Block Diagram		V1.0
Date:	Tuesday, December 17, 2024		Sheet 1 of 11

POWER DISTRIBUTION



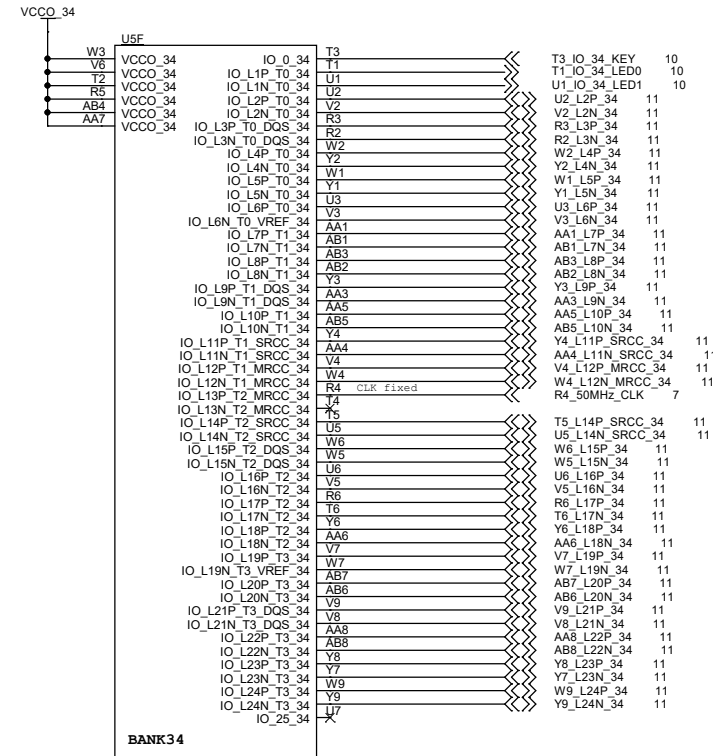
The recommended power-on sequence is
VCCINT, VCCBRAM, VCCAUX, and VCCO



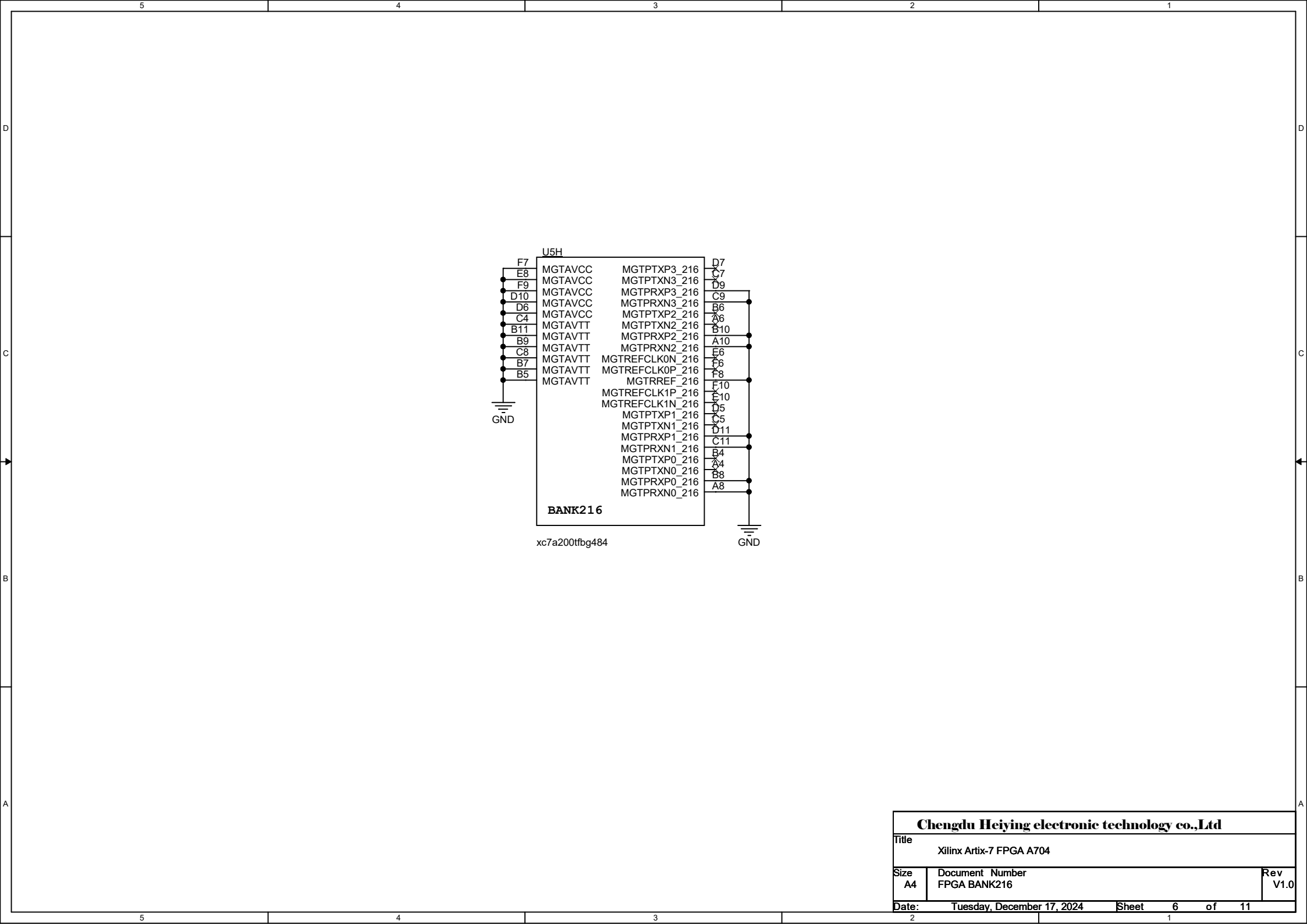


xc7a200tbg484

using the internal VREF,
the maximum data rate is 800 Mb/s (400 MHz)



xc7a200tbg484



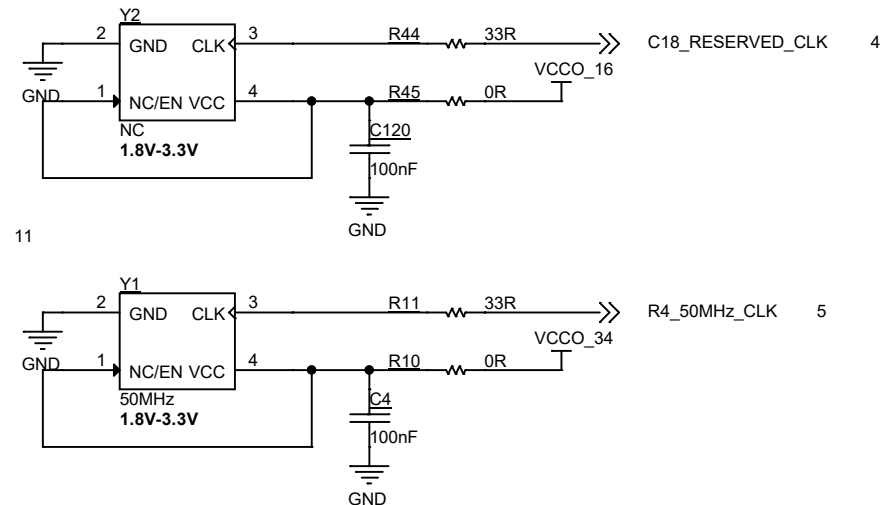
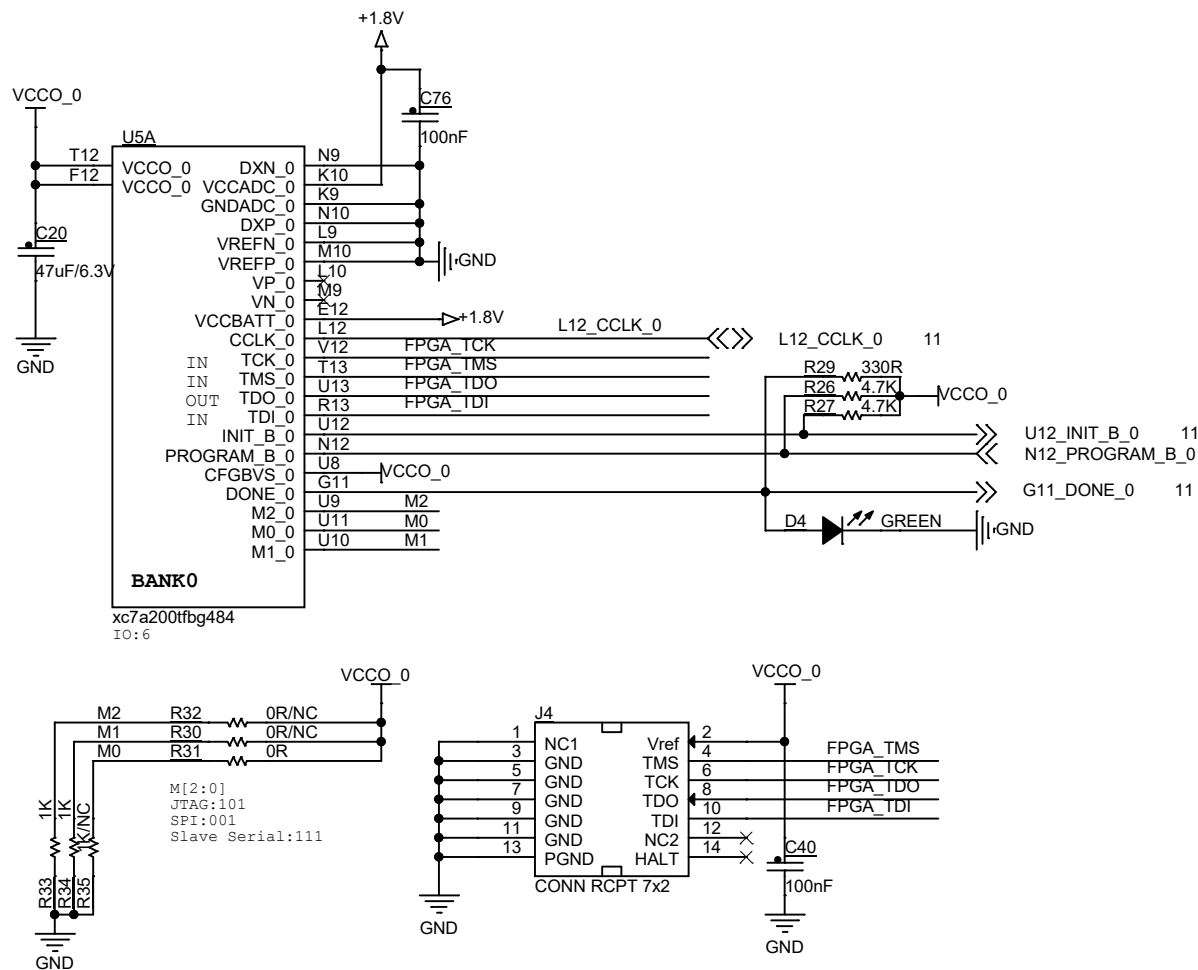
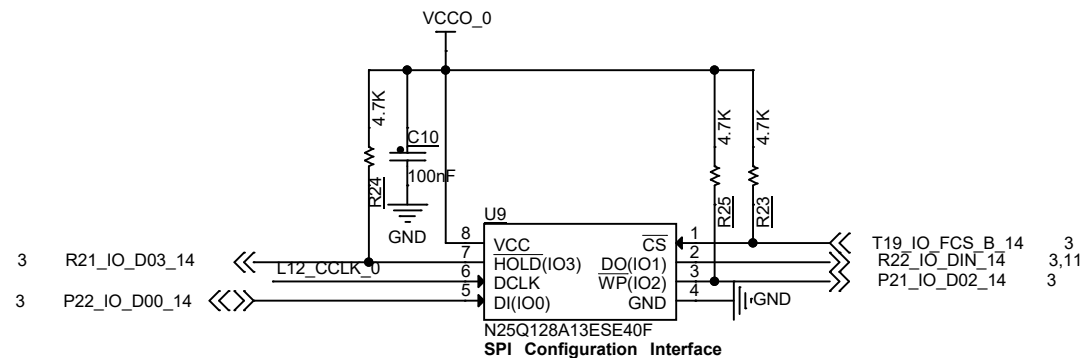


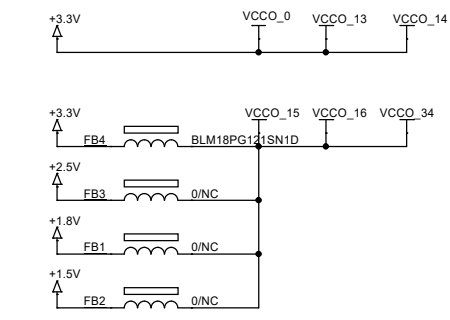
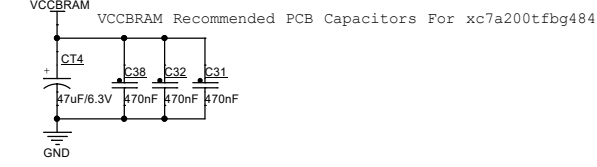
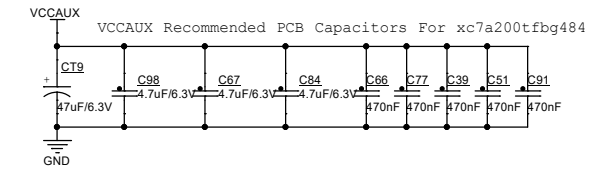
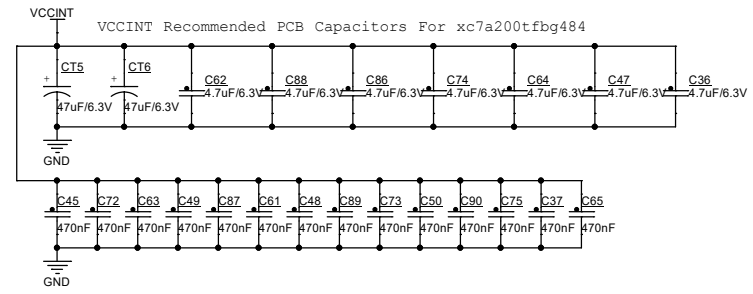
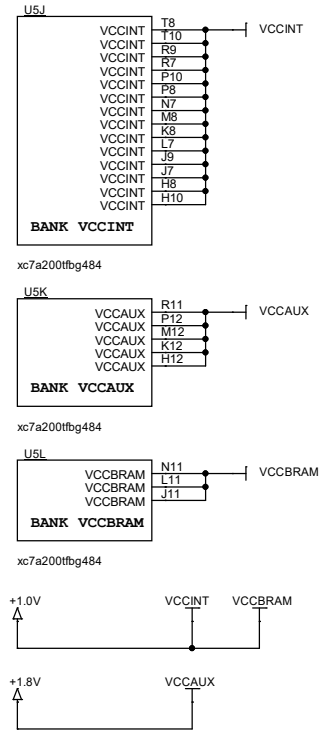
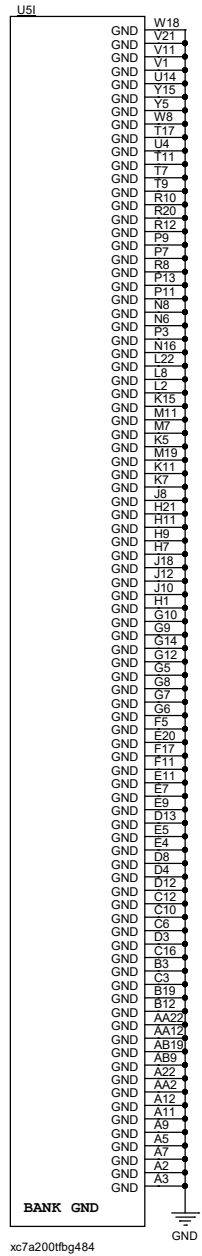
Table 2-1: 7 Series FPGA Configuration Modes

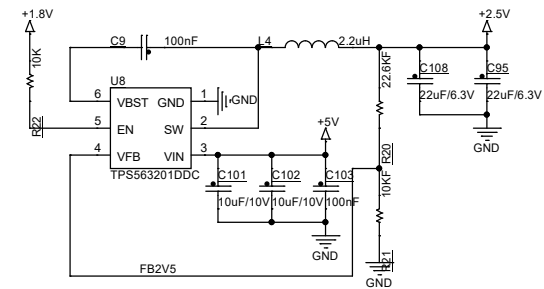
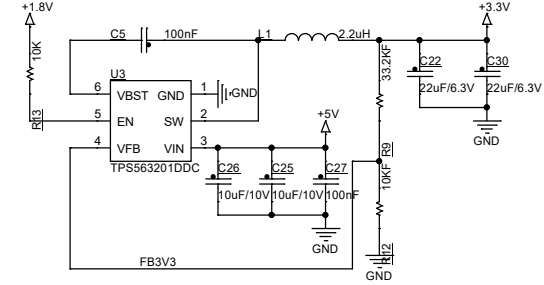
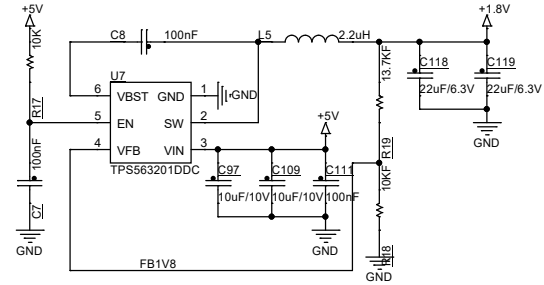
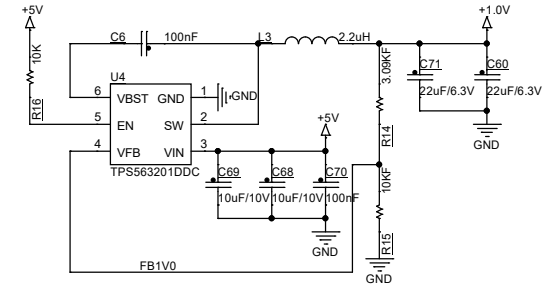
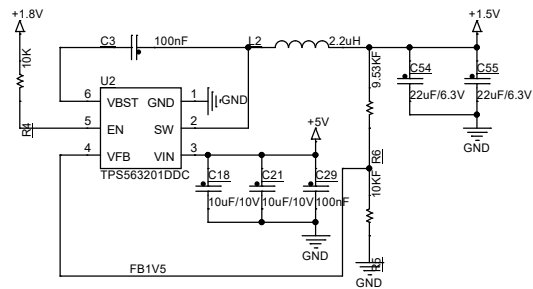
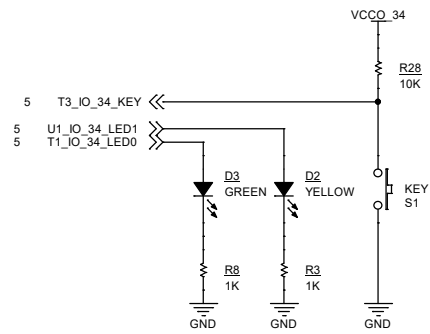
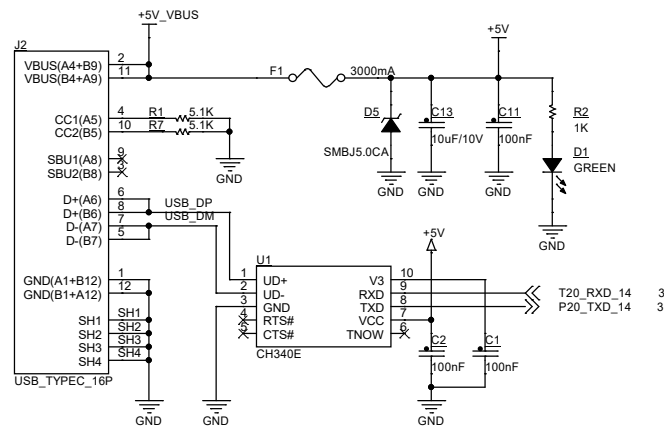
Configuration Mode	M[2:0]	Bus Width	CCLK Direction
Master Serial	000	x1	Output
Master SPI	001	x1, x2, x4	Output
Master BPI	010	x8, x16	Output
Master SelectMAP	100	x8, x16	Output
JTAG	101	x1	Not Applicable
Slave SelectMAP	110	x8, x16, x32 ⁽¹⁾	Input
Slave Serial ⁽²⁾	111	x1	Input

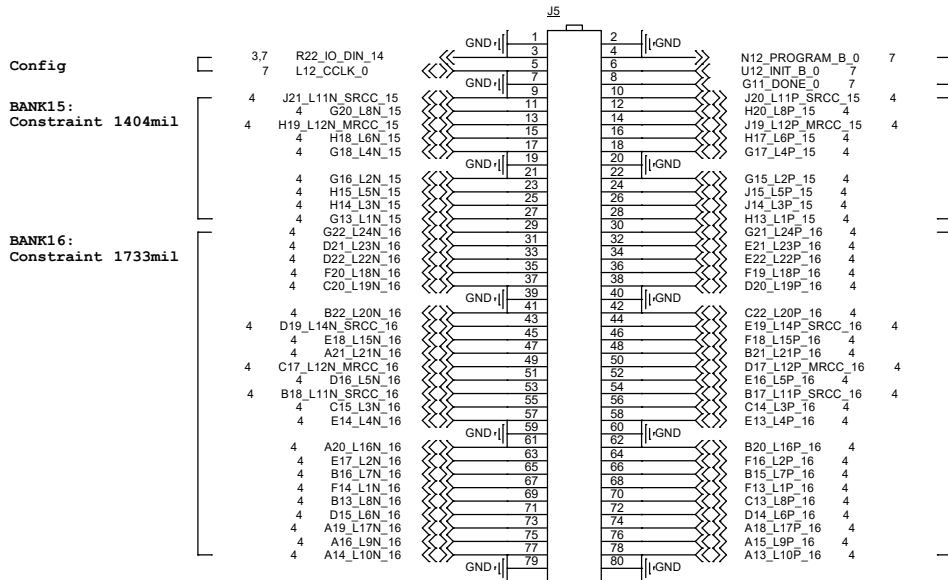
Notes:

1. The Slave SelectMAP x16 and x32 bus widths do not support AES-encrypted bitstreams.
2. This is the default setting due to internal pull-up resistors on the Mode pins.





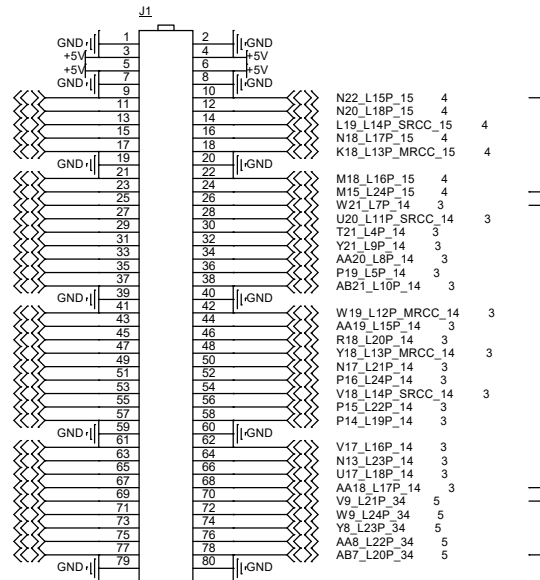
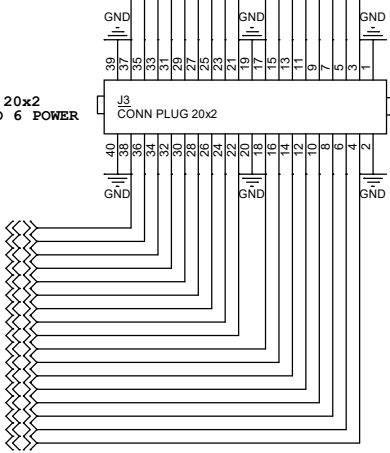




BANK34:
Constraint 1714mil

- 5 W7_L19N_34
- 5 AB5_L10N_34
- 5 AA6_L18N_34
- 5 W5_L15N_34
- 5 V5_L16N_34
- 5 AB2_L8N_34
- 5 AA4_L11N_SRCC_34
- 5 AA3_L9N_34
- 5 Y1_L5N_34
- 5 AB1_L7N_34
- 5 V3_L6N_34
- 5 W4_L12N_MRCC_34
- 5 Y2_L4N_34
- 5 T6_L17N_34
- 5 U5_L14N_SRCC_34
- 5 V2_L2N_34
- 5 R2_L3N_34

CONN 20x2
34 IO 6 POWER



BANK14:
Constraint 1606mil